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A Green, Reagent-Saving Analytical Strategy for Equivalence Point Determination in Potentiometric Titrations via Gran and Schwartz Linearization[†]

Samuele Giani,^{*a} and Cosimo A. De Caro,^{*b}

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Potentiometric titrations remain a cornerstone of analytical chemistry, yet conventional methods require titration well beyond the equivalence point, generating unnecessary chemical waste. Here we present GranTED, an open-source Python toolkit that implements an automated, two-stage Gran–Schwartz linearization strategy for significantly more reagent-efficient equivalence point determination. In the first stage, a conventional, full titration is performed past the equivalence point to establish a reliable reference equivalence point (EQP*). In the second stage, systematic backward trimming of the same dataset combined with Gran–Schwartz linearization identifies the earliest acceptable volume (V_{opt}) at which the Gran–Schwartz equivalence point (EQP) converges within a defined accuracy threshold (0.010–0.050 mL). This optimised volume is systematically smaller than both the full titration volume used and the EQP* volume as well, enabling direct reagent savings. The strategy was validated on two benchmark aqueous systems: TRIS/HCl (weak base–strong acid) and KHP/NaOH (weak acid–strong base). Depending on the chosen accuracy threshold, reagent savings of approximately 10–30% per titration were achieved while maintaining excellent agreement with the reference EQP*. These results directly support Green Chemistry Principle 1 (prevention of waste) and Principle 5 (safer solvents and auxiliaries) by reducing titrant consumption and laboratory waste without compromising analytical accuracy or reliability. GranTED is freely available, providing an accessible, reproducible, and sustainable platform for both research, quality control and teaching laboratories.

^a Universitätsspital Zürich, Rämistrasse 100, 8091 Zürich, Switzerland. E-mail: samuele.giani@usz.ch

^b Mettler-Toledo Technologies GmbH, Heuwinkelstrasse 3, 8606 Nänikon, Switzerland. E-mail: cosimo.decaro@mt.com

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Introduction

1 Introduction

Potentiometric titrations are among the most widely used analytical techniques in both research, quality control and teaching laboratories for the accurate determination of acid–base equilibria, redox processes, and complexation reactions. Conventional procedures typically require titration well beyond the equivalence point (post-EQP) to reliably identify the inflection point via derivative analysis. This practice leads to unnecessary consumption of titrant and generates avoidable chemical waste, conflicting with core principles of Green Chemistry.

Classical linearization methods, first introduced by Gunnar Gran in 1950–1952 and later extended by Schwartz in 1987, offer a powerful alternative by transforming the sigmoidal titration curve into straight lines whose intersection directly yields the equivalence point. These approaches can provide higher accuracy and precision than traditional derivative methods, particularly for weak acid–base systems or poorly defined endpoints. However, their practical implementation has remained largely manual, requiring full titration datasets and subjective selection of the linear region.

Here we introduce analytical strategy GranTED (Gran Titration Equivalence point Determination), available as an open-source Python toolkit that fully automates Gran and Schwartz linearization while incorporating a novel two-stage optimisation approach. In the first stage, a complete titration past the equivalence point establishes a high-confidence reference equivalence point (EQP*). In the second stage, systematic backward trimming of the same dataset identifies the earliest acceptable volume (V_{opt}) at which the Gran–Schwartz equivalence point (EQP) converges within a user-defined accuracy threshold. This optimised volume is systematically smaller than both the full titration volume and the volume at EQP*, enabling direct reagent savings without additional experiments or loss of analytical quality.

The performance of this strategy was rigorously evaluated on

two standard benchmark systems: TRIS/HCl (weak base–strong acid) and KHP/NaOH (weak acid–strong base). By combining automated linear-region detection, convergence diagnostics, and quantitative trimming analysis, GranTED transforms a classical analytical procedure into a greener, more efficient, repeatable and and reproducible method. This work demonstrates how modern computational tools can reduce laboratory waste while maintaining analytical reliability, in full alignment with Green Chemistry principles.

Experimental

Materials and Instrumentation

All titrations were performed using analytical-grade reagents. Tris(hydroxymethyl)aminomethane (TRIS, $\geq 99.8\%$) and potassium hydrogen phthalate (KHP, $\geq 99.95\%$) were obtained from Sigma-Aldrich and used as received. Hydrochloric acid (0.1 M certified standard) and sodium hydroxide (0.1 M certified standard) solutions were purchased from Merck. Ultra-pure water (18.2 M Ω cm) was used for all solution preparations and dilutions.

Potentiometric measurements were carried out with a Metrohm 905 Titrando automatic titrator equipped with a combined glass pH electrode (Metrohm Unitrode). The electrode was calibrated daily using commercial pH 4.00, 7.00, and 10.00 buffers. Titrant delivery was performed with a 20 mL burette (0.001 mL resolution). All experiments were conducted at 25.0 ± 0.1 °C under magnetic stirring (300 rpm). Data were recorded as volume (mL) versus potential (mV).

Titration Protocols

Two standard benchmark systems were investigated: (i) TRIS/HCl (weak base–strong acid) and (ii) KHP/NaOH (weak acid–strong base).

For TRIS/HCl titrations, 50.00 mL of approximately 0.05 M TRIS solution was titrated with 0.1 M HCl. For KHP/NaOH titrations, 50.00 mL of approximately 0.05 M KHP solution was titrated with 0.1 M NaOH. In both cases, the titration was continued well past the equivalence point (typically to 120–150 % of the expected equivalence volume) to generate the complete dataset required for reference EQP* determination. Each system was analysed in sextuplicate.

Data Analysis with GranTED

All raw titration data (volume vs. potential) were processed using the developed open-source Python toolkit GranTED (version 1.0.1). The software automatically converts potential to pH, computes both Gran (g_1) and Schwartz (g_s) functions, identifies optimal linear regions via derivative-based segmentation and R^2 maximisation, and performs the backward trimming analysis at the core of the algorithm. User-defined parameters were different EQP deviation threshold 0.010–0.050 mL. All calculations were performed on a standard laptop (Python 3.11 environment).

The same datasets were also analysed with the conventional inflection-point method (first and second derivative) using the built-in Mettler software for direct comparison.

Results and Discussion

Method Development: Earliest Acceptable Point and Gran-Schwartz Optimization

The core innovation of the analytical strategy is a two-stage procedure that combines classical Gran and Schwartz linearization with systematic backward data trimming to identify the earliest volume at which a reliable equivalence point can be obtained.

In the first stage, a conventional full titration is performed well past the equivalence point (post-EQP). The complete dataset is processed with the automated Gran-Schwartz algorithm to determine a high-confidence reference equivalence point, denoted EQP*. This value serves as the benchmark for subsequent optimisation.

In the second stage, the same experimental dataset is subjected to iterative backward trimming. Starting from the full titration and progressively removing data points from the post-EQP region, the Gran-Schwartz linearization is reapplied at each step. For each trimmed dataset the Gran-Schwartz equivalence point, denoted EQP, is calculated and compared to the reference EQP*. The difference $|EQP - EQP^*|$ is evaluated against a defined accuracy threshold (0.010–0.050 mL).

The earliest volume at which the Gran-Schwartz EQP diverges and remains outside the chosen threshold for a number of consecutive points is defined as the optimised volume, V_{opt} . This V_{opt} is systematically smaller than both the total titration volume and the volume corresponding to EQP*, while still delivering an equivalence point of acceptable accuracy.

Figure 1a) shows a representative titration curve and Figure 1b) the corresponding Gran-Schwartz plot for the full dataset and Figure 1c) and d) the trimmed dataset at V_{opt} for the TRIS/HCl system. A convergence table (Table 1) illustrates how R^2 , EP, and its standard error evolve with the number of retained data points, clearly identifying the earliest acceptable volume.

Table 1 Gran-Schwartz results for the TRIS/HCl system in the vicinity of the optimal volume V_{opt} (4.40 mL, in bold). Full titration EQP* corresponds to 5.007 mL. The selected volume meets the acceptance criteria of deviation from reference EQP* smaller than 0.010 mL) with excellent linearity and low uncertainty. Except for R^2 , all values in mL

Dispensed volume	EQP	Standard Error	R^2	Deviation from reference EQP*
4.25	4.980	0.010	0.9987	-0.027
4.30	4.987	0.010	0.9988	-0.020
4.35	4.989	0.009	0.9989	-0.018
4.40	5.002	0.005	0.9997	-0.005
4.45	5.002	0.005	0.9997	-0.005
4.50	5.007	0.004	0.9998	0.000
4.55	5.006	0.003	0.9998	-0.001

This trimming-based optimisation constitutes the key methodological advance, transforming a traditional full titration into a reagent-efficient procedure without requiring additional experiments.

Method Validation: Benchmarking against Full Titration

To validate the accuracy and reliability of the analytical approach, the Gran-Schwartz equivalence points obtained at the optimised

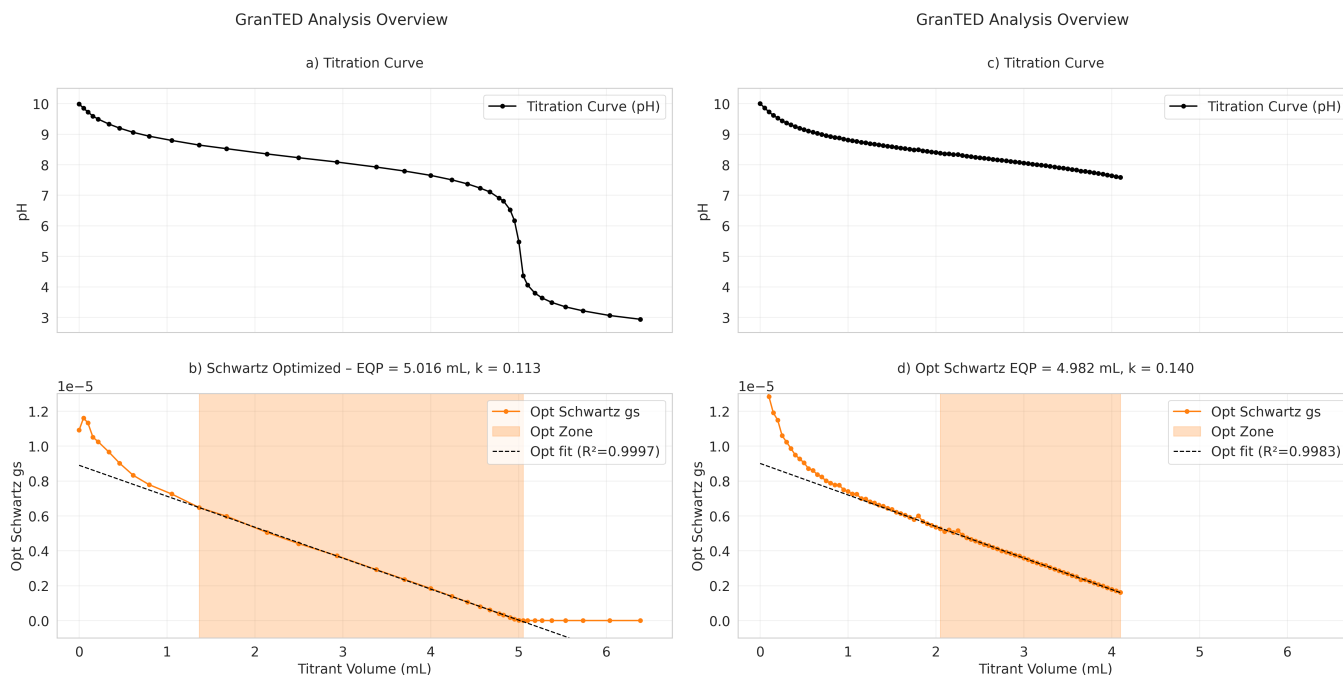


Fig. 1 Overview of the green analytical strategy for equivalence point (EQP) determination in potentiometric titrations, demonstrated for TRIS (0.1 M) titrated with HCl (0.1 M). Left: a) Raw pH vs. titrant volume (V) curve from the initial titration (full run to post-EQP). b) Schwartz–Gran modified plot ($V \times 10^{-\text{pH}}$ vs. V), showing the straight pre-EQP line extrapolated to the EQP (dashed line; intercept at 5.020 mL). Right: c) Titration performed up to the optimal earliest acceptable point volume. d) Corresponding Schwartz–Green modified plot confirming EQP extrapolation (intercept at 5.002 mL) with reduced dispensed volume.

volume V_{opt} were benchmarked against the same method applied to the full titration. Also compared is the conventional inflection-point method applied to the same full titration datasets. The inflection-point method, which relies on the maximum of the first or second derivative of the titration curve, is still widely used in laboratories but requires titration well past the equivalence point to capture the steepest slope.

For both benchmark systems, the reference equivalence point EQP* (determined from the complete post-EQP dataset) served as the ground-truth value. Table 2 summarises the results. In the TRIS/HCl system, the Gran–Schwartz EQP at V_{opt} agreed with EQP* within the accuracy of 0.010–0.050 mL (indeed the chosen threshold). The conventional inflection-point method yielded comparable agreement but only after the full titration volume was reached. The EQP in this case corresponds to 5.015 ± 0.001 mL. Similar performance was observed for the KHP/NaOH system, where the inflection-point EQP corresponds to 5.032 ± 0.001 mL.

Table 2 Comparison of equivalence point (EQP*) determination by Gran–Schwartz linear interpolation on the full titration (post-EQP) and Gran–Schwartz linear extrapolation stopped at the optimal optimised volume corresponding to the indicated deviation from reference EQP*. Results are mean $\pm$ SD ($n = 6$ replicates). All values are in mL

	Reference EQP*	Gran–Schwartz EQP		
		0.010	0.020	0.050
TRIS/HCl	5.013 ± 0.003	4.990 ± 0.010	5.005 ± 0.015	4.992 ± 0.064
KHP/NaOH	5.042 ± 0.003	5.045 ± 0.003	5.051 ± 0.015	5.066 ± 0.041

Importantly, the standard error (Table 1) and standard deviation (Table 2) associated with the Gran–Schwartz EQP remained low and comparable to that of the full method reference EQP*, demonstrating that the early-stop strategy does not compromise precision. The comparison of the equivalence points obtained by the three approaches (reference EQP*, Gran–Schwartz at V_{opt} and conventional inflection point) across both systems confirms excellent consistency.

These results demonstrate that GranTED delivers equivalence points matching analytical quality to the classical inflection-point method, yet systematically achieves this with a substantially reduced reagent volume. The automated linear-region identification and trimming algorithm eliminate the possible ambiguity inherent in derivative-based analysis while enabling reliable early termination of the titration.

Green Chemistry Metrics and Sustainability Benefits

The optimised-volume strategy implemented of the propose analytical strategy directly translates into measurable reductions in chemical consumption. For the TRIS/HCl (aq) and KHP/NaOH (aq) benchmark systems, reagent savings ranged from approximately 10% (at the strictest 0.010 mL accuracy threshold) to 30% (at the 0.050 mL threshold) relative to a conventional full titration past the equivalence point. These savings were consistently achieved across both systems, as illustrated in Fig. 2 (reagent-saving histogram).

By terminating the titration at the optimised volume V_{opt} , the volume of titrant required is reduced while the determined equiv-

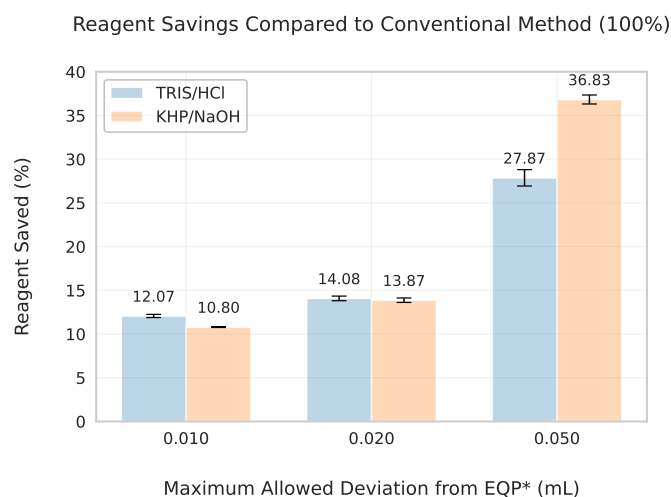


Fig. 2 Reagent savings achieved by stopping at the optimal volume using different acceptance criteria (0.010, 0.020 and 0.050 mL difference from the full-dataset EQP). Results are shown for both TRIS/HCl and KHP/NaOH systems ($n = 6$ replicates each) and compared against the equivalence point and full titration endpoint

alence point remains within the specified accuracy window. This approach avoids the unnecessary addition of excess titrant that is typical in conventional potentiometric procedures, which must extend well beyond the equivalence point to reliably identify the inflection.

The demonstrated reagent savings align directly with two core principles of Green Chemistry: Principle 1 (prevention of waste) and Principle 5 (design of safer solvents and auxiliaries). Reducing titrant consumption lowers both the amount of hazard of reagents used and the volume of waste generated per analysis. Furthermore, the open-source nature of GranTED enables widespread adoption in academic and industrial laboratories, promoting more sustainable analytical practices without requiring expensive commercial software.

Overall, the analytical strategy GranTED demonstrates that modern computational linearization combined with systematic data trimming can transform a classical analytical method into a greener, more efficient procedure while maintaining high analytical performance.

Conclusions

We have introduced a new, green analytical strategy developed within GranTED, an open-source Python toolkit that implements an automated two-stage Gran–Schwartz linearization strategy for potentiometric titrations. By combining a full titration to establish a reference equivalence point (EQP*) with systematic backward trimming, the software reliably identifies the earliest acceptable volume V_{opt} at which the Gran–Schwartz equivalence point (EQP) converges within a user-defined accuracy threshold (typically 0.010–0.050 mL).

Applied to the benchmark systems TRIS/HCl and KHP/NaOH, this approach consistently delivered equivalence points in excellent agreement with both the reference EQP* and the conventional inflection-point method, while reducing titrant consump-

tion by 10–30% per titration. The strategy requires no additional experiments or hardware and eliminates the subjectivity associated with manual linear-region selection.

GranTED therefore provides a practical, accurate, and significantly greener alternative to traditional potentiometric titration workflows. By enabling earlier termination of titrations without loss of analytical quality, it directly supports Green Chemistry Principle 1 (prevention of waste) and Principle 5 (design of safer solvents and auxiliaries). As a freely available, MIT-licensed tool with comprehensive diagnostic and visualisation capabilities, GranTED is expected to find wide application in research, quality control, and teaching laboratories seeking more sustainable analytical practices.

Future extensions will include support for complexometric, precipitation, and redox titrations, as well as batch processing for high-throughput green analytical workflows.

Author Contributions

Author Contributions

S.G. and C.A.D.C. conceptualized the study, developed the methodology, performed the investigation and formal analysis, curated the data, visualized the results, and wrote the original draft. S.G. developed the software. C.A.D.C. carried out the experiments.

Conflicts of Interest

S.G. declares no conflicts of interest. C.A.D.C. is currently working for a manufacturer of laboratory analytical instrumentation (Mettler-Toledo Technologies GmbH).

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Data Availability

The datasets supporting this article are available in the Electronic Supporting Information (ESI) or from the corresponding authors upon reasonable request.